

Temperature-Aware Farm System

Fundamentals of AI Using Agriculture Data Set

Student Details

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1. Introduction

Agriculture plays a vital role in food security and economic development. Farmers often suffer losses due to crop spoilage caused by high temperatures and inadequate storage facilities. Artificial Intelligence can help predict spoilage risks and support timely decision-making.

This project develops a Temperature-Aware Farm System that predicts spoilage risk using machine learning techniques.

2. Problem Statement

High temperatures and poor storage conditions can lead to significant crop spoilage. Farmers need an intelligent system that can identify spoilage risks before losses occur.

3. Objectives

- Analyze agricultural data related to temperature and storage.
- Develop a machine learning model to predict spoilage.
- Identify important factors affecting spoilage.
- Assist farmers in making informed decisions.

4. Dataset Description

The dataset contains the following attributes:

- Record_ID
- Date
- Location
- Crop_Type
- Temperature_C
- Storage_Available
- Spoilage_Observed

Additional features were generated:

- Relative_Humidity_%
- Heatwave_Alert
- Days_In_Storage
- Transport_Delay_Hours

5. Methodology

Data Preprocessing

- Data loading using Pandas
- Feature engineering
- Label encoding of categorical variables

Machine Learning Model

- Supervised Learning
- Random Forest Classifier

Evaluation

- Accuracy Score
- Confusion Matrix
- Feature Importance Analysis

6. Results

The Random Forest model successfully predicts spoilage risk using temperature, storage conditions, and other environmental factors.

Feature importance analysis showed that temperature and storage-related factors significantly influence spoilage prediction.

Model Accuracy

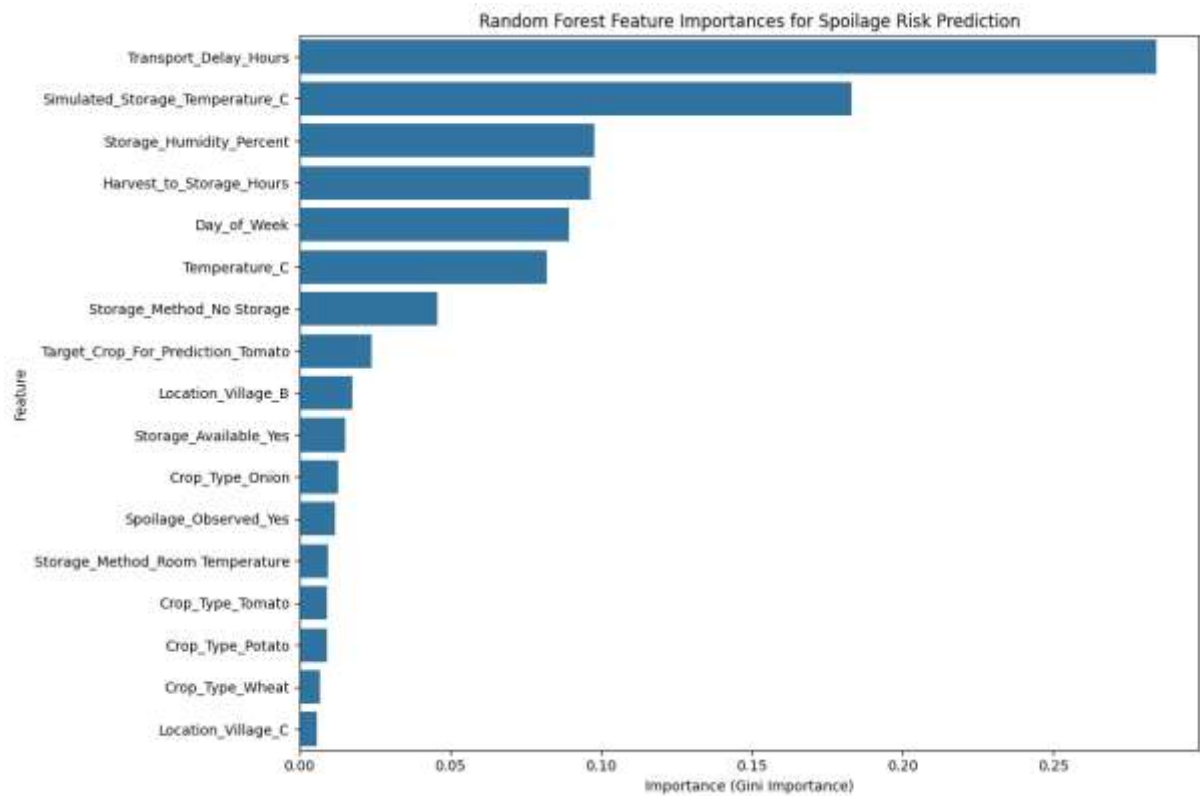
... Random Forest Classifier Performance (n_estimators=50):					
Accuracy: 0.9000					
Classification Report:					
	precision	recall	f1-score	support	
Low	0.00	0.00	0.00	0	
Medium	0.00	0.00	0.00	2	
High	0.90	1.00	0.95	18	
accuracy			0.90	20	
macro avg	0.30	0.33	0.32	20	
weighted avg	0.81	0.90	0.85	20	

Confusion Matrix

Confusion Matrix Values (Labels: 0=Low, 1=Medium, 2=High):

```
[[ 0  0  0]
 [ 0  0  2]
 [ 0  0 18]]
```

Feature Importance Graph



7. Benefits

- Early spoilage prediction
- Reduced agricultural losses
- Better storage management
- Improved decision-making for farmers

8. Future Scope

Future improvements may include:

- IoT sensor integration
- Real-time weather forecasting
- Mobile application support
- Automated farmer alerts

9. Conclusion

The Temperature-Aware Farm System demonstrates how Artificial Intelligence can be applied in agriculture to reduce spoilage losses and improve farm productivity. The project highlights the effectiveness of machine learning in supporting agricultural decision-making.

10. What Did This Case Study Help Me Infer?

This case study helped me understand how Artificial Intelligence can be applied to real-world agricultural challenges. By analyzing temperature, storage conditions, and spoilage-related factors, I learned how data can be transformed into actionable insights that support farmers in reducing losses and improving productivity.

The project demonstrated that temperature and storage conditions have a significant impact on crop spoilage. Through machine learning techniques, it is possible to identify risks early and take preventive actions before spoilage occurs.

Overall, this case study highlighted the importance of AI-driven agricultural systems and their potential to improve decision-making, resource management, and food security.

11. Blog Article: Temperature-Aware Farm System Using AI

Agriculture is highly dependent on environmental conditions, especially temperature. Sudden increases in temperature can affect crop quality, increase spoilage risk, and reduce profitability for farmers. Traditional monitoring methods may not always provide timely warnings, making Artificial Intelligence an effective solution.

In this project, a Temperature-Aware Farm System was developed using agricultural data and machine learning techniques. The system analyzes factors such as temperature, storage availability, humidity, transport delays, and heatwave alerts to predict spoilage risk.

The analysis showed that temperature and storage-related factors are among the most important contributors to spoilage. By identifying these risks in advance, farmers can take preventive measures such as improving storage conditions, adjusting transportation schedules, and monitoring environmental changes more effectively.

This project demonstrates how AI can support sustainable agriculture by reducing losses, improving productivity, and enabling data-driven decision-making.

12. Can This Be a GitHub Repository?

Yes. This project is suitable for a GitHub repository because it contains:

- Dataset analysis and preprocessing
- Machine learning model implementation
- Data visualization and output results
- Project report and documentation
- Agricultural AI application use case

Publishing the project on GitHub allows collaboration, learning, and future improvements by students, researchers, and developers.

13. Better Solution and Future Improvements

Future improvements may include:

- Integration with IoT-based temperature sensors.
- Real-time weather forecasting support.
- Mobile application for farmers.
- Automated spoilage alerts and notifications.
- Inclusion of humidity, rainfall, and soil data.
- Advanced machine learning and deep learning models.

14. Final Thoughts

This case study inspired me to think beyond theoretical learning and focus on solving practical agricultural problems. It demonstrated how Artificial Intelligence can be used to create solutions that directly benefit farmers and improve agricultural sustainability.

The project encouraged innovation, problem-solving, and the application of technology to address real-world challenges. It also highlighted the importance of using AI responsibly to support communities and improve quality of life.

References

1. Python Documentation
2. Scikit-Learn Documentation
3. Pandas Documentation
4. IIT Ropar ANNAM.AI Course Material